Unsupervised ECG Anomaly Detection with an MLP Autoencoder

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Course: Machine Learning Mini-Project | Code: https://github.com/valelobo/EGC-Anomaly-Detection.git

1 Problem and Motivation

Cardiovascular diseases are the leading cause of death worldwide, at about 17.9 million deaths per year [1]. The electrocardiogram (ECG) is the standard non-invasive test for heart rhythm disorders. Traditionally, a cardiologist inspects it visually, either as a short 12-lead recording or as a 24–48 h Holter recording. A single day of Holter monitoring contains around 100 000 heartbeats, so manual review is slow, tiring and prone to error. Rare abnormal beats are easily missed, and they may be early signs of a serious condition such as ventricular arrhythmia or heart failure.

This project automates the screening step: each heartbeat is scored by how much it deviates from normal morphology, and suspicious beats are flagged for clinical review. The aim is to support the cardiologist, not replace them, by directing expert attention to where it is needed. **Input:** one heartbeat, a time series of 140 samples. **Output:** an anomaly score (the reconstruction error) and a label, *normal* or *anomaly*.

2 Dataset and Method

Data. The ECG5000 dataset [2, 3] contains 4998 heartbeats from a 20-hour recording in the BIDMC Congestive Heart Failure Database. Each beat was segmented and interpolated to a length of 140 samples. The copy hosted by TensorFlow [4] merges the original five classes into normal (2919 beats) and anomalous (2079 beats). The data were split 70/15/15 into training, validation and test sets, stratified by class, with seed 42. Signals were min-max scaled using statistics from the training data only.

Model. The model is a fully connected (MLP) autoencoder with layers $140 \to 64 \to 32 \to 8 \to 32 \to 64 \to 140$, ReLU activations and a sigmoid output. It was implemented in PyTorch [5]. An autoencoder learns in a *self-supervised* way, because the input itself is the target. The 8-unit bottleneck forces the network to learn a compact representation of the training signals [6]. The network was trained *only on normal beats* for 100 epochs, using L1 loss, Adam (learning rate 10^{-3}) and a batch size of 64. The weights with the lowest validation loss on normal beats were kept. This makes the detector a *one-class*, unsupervised method [7, 8]: it learns what "normal" looks like without labelled examples of every disease, a key advantage because anomalies are rare, varied and expensive to annotate.

Decision rule and evaluation. The anomaly score of a beat $\mathbf{x}$ is its mean absolute reconstruction error, $e = \frac{1}{140}\sum_t |x_t - \hat{x}_t|$. A beat is flagged when $e > \tau$. The threshold τ was chosen to maximise F1 on the validation set, and all reported metrics are computed on the held-out test set. The two types of error carry very different risks. A *false negative* is a missed anomaly, which may delay the diagnosis of a potentially fatal condition. A *false positive* is a false alarm, which leads to unnecessary tests, patient anxiety and alarm fatigue in clinical staff. Accuracy alone can hide these errors, so the confusion matrix, recall (sensitivity), specificity, precision and F1-score are reported, together with the threshold-independent AUROC and AUPRC.

3 Results and Discussion

Table 1 summarises the test results. An AUROC of 0.980 shows that the reconstruction error separates normal and anomalous beats almost perfectly. At the chosen threshold the detector finds 96.5% of anomalies (recall) and correctly passes 95.9% of normal beats (specificity). This gives an F1-score of 0.954. Out of 750 test beats, only 11 anomalies were missed and 18 normal beats were falsely flagged.

Table 1: Test-set performance (750 beats, 312 anomalies) at threshold $\tau = 0.035$.

AUROC	AUPRC	Accuracy	Precision	Recall	Specificity	F1
0.980	0.954	0.961	0.944	0.965	0.959	0.954

Figure 1 shows why the method works. Normal beats are reconstructed with low error, while most anomalous beats form a separate group with errors 3–4 times higher. The few errors lie where the two distributions overlap. Figure 2 shows the mechanism. The autoencoder reproduces a normal beat closely. For an anomalous beat, its output is pulled towards the normal shape it has learned, so the error is concentrated where the beat departs from normal morphology. The error therefore also shows *where* a beat is abnormal.

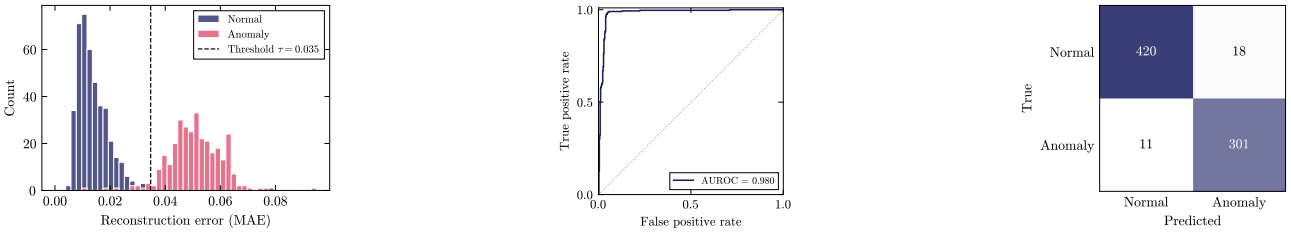


Figure 1: Test set. Left: reconstruction-error distributions and the threshold. Centre: ROC curve. Right: confusion matrix.

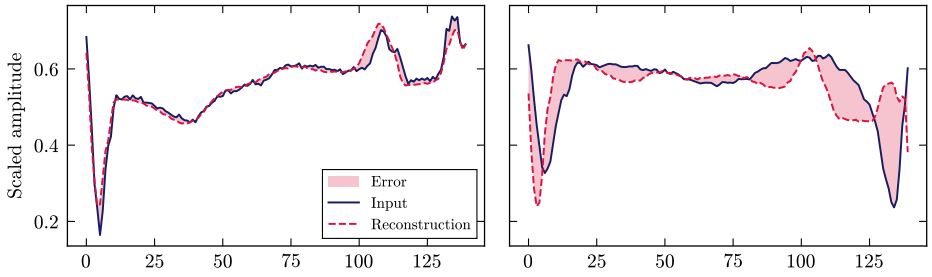


Figure 2: Input beat (navy), autoencoder reconstruction (red, dashed) and reconstruction error (shaded) for (a) a normal and (b) an anomalous test beat.

4 Conclusions and Reflection

A simple MLP autoencoder trained only on normal heartbeats detects anomalous beats with an AUROC of 0.980 and an F1-score of 0.954, a strong and inexpensive baseline for ECG screening. The main lessons concern handling signals and evaluating models trained on them. All preprocessing statistics must be fitted on training data only, the threshold must be chosen on a validation set rather than on the test set, and the metrics must reflect the real cost of each error type, not just accuracy.

Limitations. ECG5000 comes from a single patient and contains clean, pre-segmented beats, so performance on other patients and on raw, noisy recordings is not demonstrated. The threshold also used labelled validation anomalies. A fully label-free alternative would be a percentile of the reconstruction errors of normal beats. Possible improvements are training on multi-patient data (for example, the MIT-BIH database) and using convolutional or recurrent autoencoders that exploit temporal structure.

Relevance to my field. The same framework applies directly to structural health monitoring of infrastructure. There, sensor signals such as vibrations or strains are plentiful for the healthy state, but examples of damage are rare or unavailable. A model trained on the healthy baseline can flag deviations as possible damage. As in the clinical case, a missed defect (false negative) can be critical for safety, while frequent false alarms erode trust in the monitoring system. The evaluation approach used here therefore transfers directly.

Reproducibility and Disclosures

Code and instructions are in `script.py` and `README.md`; `python script.py train` downloads the public data and reproduces all results (seed 42). No pretrained models were used. Generative AI (Claude, Anthropic) assisted with the charts and english spelling corrections, all results were checked by the author.

References

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